

Exercise 09: Cryptography

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Duration

30 minutes for hands-on exercise + **1 hour** for writing report

Exercise 1

1. The following text was encrypted with the Caesar cipher. Decrypt it (The key is not 3).

IWXH XH P KTGN DAS RXEWTG

2. Challenge: decrypt the following text that was encoded with a substitution cipher.

Wpmxdl gfa efhza hs b wpzz, wbmc, bdw ihpdwzaii wbq xd gfa bpgptd hs gfa qabm, efad gfa kzhpwi fpdl hjjmaiixvazq zhe xd gfa fabvadi, X fbw yaad jbiixdl bzhda, hd flmiaybkc, gfmhplf b ixdlpzbmzq wmabmq gmbkg hs khpdgmq; bdw bg zadlgf shpdw tqiazs, bi gfa ifbwai hs gfa avadxdl wmae hd, exgfd vxae hs gfa tazbdkfhzq Fhpia hs Pifam. X cdhe dhg fhe xg ebi – ypg, exgf gfa sxmig lzxtjia hs gfa ypxzwxdl, b iadia hs xdipssambyza lzhht jamvb wdw tq ijxmxg. X ibq xdipssambyza; shm gfa saazxdl ebi pdmazxavaw yq bdq hs gfbg fbzs-jzabipmbyza, yakbpia jhagxk, iadgxtadg, exgf efxfk gfa txdw pipbzzq makaxvai avad gfa igamdaig dbgpmbz xtblai hs gfa waihznga hm gammxyza.

Exercise 2

1. Argue which is efficient/faster/computation-costly AES or RSA.
2. Argue why AES is not broken yet.
3. Argue about the difference between RSA encryption and ElGamal encryption schemes.

Exercise 3

1. In RSA, assume that $p = 5, q = 11$ as in the lecture, and $e = 13$. What is the partial secret key d ? When a plaintext is $m = 9$, what is the ciphertext c ? Decrypt the c to check if it is correctly decrypted to m .
2. In ElGamal, assume that $p = 23, g = 5$ as in the lecture. Pick x as you like and compute h . When a plaintext is $m = 10$, what is (c_1, c_2) ? Note that you can freely choose randomness k . Decrypt to check the correctness.
3. What kind of smart idea do you need to make RSA secure? Hint: RSA-OAEP.

Exercise 4

Imagine you want to brute-force an 80-bit key. Assume that your computer is able to try out 2^{30} keys per second.

- How long will it take your computer to try all keys?
- Do you think this is a conservative or optimistic estimate?

Exercise 5

Visit your favourite website. Determine the following:

1. Is the connection encrypted?
2. What version of TLS is used?
3. What algorithms are used? Can you figure out what kind of cryptographic algorithms those are?